

PRODUCT DOCUMENT

A Generative AI Framework for Automatic Comic Strip Generation – Product Specification Document

1. Product Overview

The AI Comic Generator is an end-to-end generative system that transforms user-written scenarios into complete comic strips using multimodal AI. It integrates LLM-based script generation, image diffusion models, and automated layout rendering to produce coherent, image-rich comic panels suitable for entertainment, storytelling, education, and rapid visual prototyping.

2. Product Purpose

The goal of the product is to:

- Automate comic creation
 - Reduce dependence on artistic skills
 - Provide a fast, accessible creative tool
 - Enable widespread usage in storytelling, social media, and content generation
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3. Core Functionalities

1. Scenario Interpretation

The system uses an LLM to parse the user's input scenario, generating:

- Number of panels
- Scene descriptions
- Dialogue
- Action flow

2. Image Generation

A text-to-image model generates comic-style illustrations for each panel with:

- Fixed resolution (512×512)
- Stylized appearance
- Coherent scene depiction

3. Text Integration

The system automatically:

- Renders dialogue onto separate text images
- Merges text with the artwork beneath each panel

4. Comic Strip Assembly

A layout engine:

- Normalizes panel size
- Arranges them horizontally
- Outputs a final stitched comic strip

5. User Interface

A Gradio-based UI provides:

- Input fields
 - Action buttons
 - Output galleries
 - Real-time status updates
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4. Architecture Summary

Input → Script Generation → Image Generation → Text Rendering → Panel Stitching → Output

Components:

- **Frontend:** Gradio interface
 - **LLM:** OpenAI (GPT-4.1-mini or similar)
 - **Diffusion Model:** Stability AI gRPC pipeline
 - **Backend Logic:** Python, PIL, JSON parser
 - **Output Renderer:** PIL image processing
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5. Product Features

Functional Features

- Automatic story breakdown into panels
- Comic-style image generation
- Panel text rendering
- Comic-strip output
- Interactive UI

Non-Functional Features

- Requires API authentication
- Real-time generation
- Extensible architecture

- Lightweight computation (cloud hosted)
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6. Target Users

- Students & educators
 - Writers & storytellers
 - Social media content creators
 - Hobbyists
 - Comic designers (rapid prototyping)
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8. Product Advantages

- Fully automated
 - No technical or artistic skills required
 - Works for any type of scenario
 - Fast generation (5–15 seconds per panel)
 - Clean UI
 - Easily deployable on cloud hosting platforms
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9. Deployment Overview

The product can be hosted on:

- Hugging Face Spaces
- GitHub Codespaces

- Render
 - Replit
 - Local machine using `demo.launch()`
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10. Compliance & Safety

- Only uses APIs approved for generative media
- All outputs depend on user text and model filters
- Content blocked by Stability AI's safety filter is handled safely